GrowthTrack

HacXLerate 2026 Grand Finale: detailed slide plan

Format: 7 slides maximum, 7 minutes presenting, 3 minutes jury Q&A, no live demo needed.
Reporting: 11 AM at the byteXL office.

SCORING WEIGHTS AND HOW THIS DECK ANSWERS THEM

Criteria	Weight	Where this deck earns it
Innovation & solution approach	35%	Slides 3, 4: predict, explain, recommend, what-if, with an owner on every action
Problem & business understanding	30%	Slides 1, 2, 6: four roles, real decisions, eligibility rules, outcomes that matter
Presentation & communication	15%	One message per slide, split speaking, rehearsed timing
Jury Q&A / defence	20%	Slide 5 evidence, honest limits, Q&A sheet at the end

TIME PLAN (7:00)

Slide	Title	Time	Suggested speaker
1	The problem and the stakes	1:00	Member A
2	User, business and key insight	1:00	Member A
3	Our solution and what is innovative	1:15	Member B
4	Approach and architecture	1:15	Member B
5	Why this approach works	1:00	Member C
6	Impact, feasibility and next steps	0:45	Member C
7	Why our solution should win	0:30	Member A

THE ONE MESSAGE TO REPEAT

> **GrowthTrack turns student data into predictions, then into specific actions that a named person takes.**

RULES FOR EVERY SLIDE

- Headline is a full sentence. Maximum about 25 words of body text; use diagrams and numbers.
- Risk colours stay constant: red = High, amber = Medium, green = Low.
- Every number on a slide comes from the project (shown in the "Source" line of each page).
- Say the data is synthetic yourselves, early, so the jury hears it from you.
- Everyone must be able to explain every slide; speaking parts are split only for delivery.

Do not claim: production login is live (demo auto-signs-in by role area; enforcement exists but is off), that every page reads the database (some shared admin pages still read the bundled dataset), or that engagement-drop prediction is reliable.

Slide 1: The Problem & The Stakes

1:00 · Member A

> **Student risk is found late, and one score cannot see all of it.**

ON THE SLIDE

- **Left, the situation:** attendance, marks, LMS activity, placement tests, skills, engagement and faculty feedback live in separate places.
- **Centre, the stakes:** four outcomes decide a student's path: SGPA below 6, a new backlog, placement readiness before the season, an engagement drop.
- **Right, the gap:** the Round 1 Success Score blends seven areas into one number.

Round 1 Success Score weight	Academic	Attendance	LMS	Placement	Skills	Engagement	Faculty
Weight	.25	.20	.15	.15	.10	.10	.05

- **The big number:** that score predicts engagement drop with AUC **0.44** (worse than useful) while it reaches 0.77 to 0.85 on the other outcomes.

VISUAL

A horizontal timeline: "Result declared / backlog appears" on the right, "Warning signs" (falling attendance, fewer LMS logins, late submissions) far earlier on the left, with a gap labelled "intervention window lost today". Beside it, one big "0.44" with the caption "Round 1 score vs engagement drop".

SPEAKER SCRIPT (ABOUT 130 WORDS)

> *"A college holds a lot of data about each student, but it sits in different places. By the time a semester result shows a backlog, the chance to help has gone. Our Round 1 platform combined seven areas into one Success Score. It worked reasonably for marks and placement, but for engagement drop it scored 0.44, which means it was not useful at all. Mentors need to know who will struggle, why, and what to do before it happens. That is the problem we solved in Round 2."*

TRANSITION

"So who needs this, and what decision do they face?"

AVOID

- Invented statistics about dropouts or mentor workloads: the project does not contain them.

Source: student-portal/lib/scoring.ts weights; data/DATASET.md and model-metrics.json (Success Score proxy AUC).

Slide 2: User, Business & Key Insight

1:00 · Member A

> **Four people make decisions with student data; each needs a different answer.**

ON THE SLIDE: FOUR ROLE CARDS

Role	Question they ask	What GrowthTrack gives them
Faculty mentor	Which of my mentees needs attention first?	Mentee-only view, predictions with drivers, prioritised recommendations
HOD	How is my department doing?	Department results and attendance, faculty view, and the control to lock or unlock the student feedback window
TPO	Who is placement-ready and who needs support?	Placement analytics, readiness, students needing support, strong and weak areas
Student	Where do I stand and what do I do now?	Home plan with top actions, attendance planner (how many classes can I miss), results, placement eligibility, skills, coding and interview practice

Admin also manages attendance sessions, declared results, interventions and settings. Sees everything.

KEY INSIGHTS (THREE, ONE LINE EACH)

- **1. Good CGPA is not employability.** A student can have strong grades and weak aptitude, coding or interview scores.
- **2. Some students fall silently.** The "silent decliners" segment (27 students) is invisible in grades alone.
- **3. A prediction is useless without an action.** Every recommendation needs a reason, an owner and a priority.

BUSINESS VALUE

- Fewer backlogs and better placement readiness by acting before results.
- Student-facing rules are explicit: eligibility needs CGPA at least 7, no active backlog, attendance at least 75%.

VISUAL

Four equal cards, one icon each, with the question in bold. Bottom strip with the three insights as short phrases.

SPEAKER SCRIPT (ABOUT 130 WORDS)

> *"We designed for four people. A faculty mentor asks who needs attention first. A head of department wants the department picture and controls such as opening the feedback window. The placement officer needs to know who is ready and who needs support. And the student needs a clear plan. Three insights drive the design: good marks do not mean employability, some students decline silently, and a prediction only matters if someone owns the action."*

Source: role workspaces (app/hod, app/tpo, app/faculty, student portal); data on segments (public/data/segments.json); placement eligibility in scoring.ts.

Slide 3: Our Solution & What Makes It Innovative

1:15 · Member B

> **One product that predicts, explains, recommends and lets you test changes.**

ON THE SLIDE: THE LOOP

PREDICT (4 outcomes, risk level) → **EXPLAIN** (per-student drivers) → **RECOMMEND** (action + owner + priority) → **WHAT-IF** (change an input, re-score) → back to the mentor's decision.

THE FIVE INNOVATION POINTS (ONE LINE EACH ON THE SLIDE)

Feature	What it is in the project	Why it matters
Four predictions	SGPA below 6, backlog risk, placement readiness, engagement drop, each with probability and Low/Medium/High	Covers academics, careers and engagement together
Explainable drivers	Per-student reasons built from summed feature families, filtered so they never contradict common sense	Mentors can trust and act on them
Recommendation engine	1,027 recommendations across 594 students plus 11 segment-level; each has action, reason, owner (faculty mentor, T&P cell or counsellor) and priority; 6 segments	Specific, owned, prioritised, not generic advice
What-if simulator	Edit attendance this semester, LMS logins per month, assignments submitted %, best mock interview score, best coding score; the student is re-scored in the browser and matches the Python model	Shows the student what to change
Analytics copilot	Natural-language questions answered from the platform database	Leaders ask in plain language

ALSO BUILT (ONE MENTION EACH, DO NOT LINGER)

- **Strict mock-interview scoring:** checks relevance before scoring; irrelevant, copied, gibberish or "I don't know" answers get 0; keyword stuffing is penalised; scored on the server.
- **Executive brief page** for the Dean; **HOD feedback lock**; **anonymous faculty evaluation**.

VISUAL

A circular loop diagram with the four steps, plus a small screenshot strip: prediction with drivers, recommendation list, what-if sliders.

SPEAKER SCRIPT (ABOUT 160 WORDS)

> *"GrowthTrack is a loop. First we predict four outcomes with a risk level. Then we explain why, in plain terms, per student. Then we recommend: every recommendation names the action, the reason, the owner and the priority, and we generate over a thousand of them, so mentors are not given generic advice. Finally, the what-if simulator lets a student or mentor change attendance, LMS activity, assignments, or practice scores, and see the risk re-calculated instantly. On top we have an analytics copilot, and a mock-interview scorer that refuses to reward irrelevant answers."*

Source: ml/, lib/recommendations.ts, lib/whatif.ts, public/data/whatif-model.json editable inputs, evaluation/.

Slide 4: Approach & Architecture

1:15 · Member B

> **A leakage-safe model pipeline feeding role-based dashboards on a secured database.**

ON THE SLIDE: LEFT-TO-RIGHT DIAGRAM

Data (800 students, 5 departments, 15 tables) → **Features** (only data up to end of semester N) → **Models** (logistic regression; gradient boosting only if it wins by 0.01 AUC) → **Calibrated risk + drivers** → **Recommendation rules + priority** → **Role dashboards, copilot, what-if**

MODEL PIPELINE

- Python with scikit-learn.
- Time-based feature cut *before* imputing, to avoid leakage.
- Recall tuned to 0.85 in cross-validation.
- Two-tier risk: Medium at the recall threshold, High at a stricter level, so mentors are not flooded.
- Calibrated probabilities; train/test split of 640 / 160.
- Same logistic models rescored in TypeScript, matching Python.

PLATFORM

- Next.js 16, React 19, TypeScript, Tailwind v4, recharts.
- Supabase Postgres + Auth; 45 tables.
- Row-level security on every table; roles: admin, faculty, HOD, TPO, student.
- Mentor sees mentees, HOD sees a department, TPO sees all.
- Anonymous faculty evaluations; students cannot edit marks or predictions.

STUDENT PORTAL AND ADMIN FEATURES (THE PRODUCT AROUND THE MODELS)

- Student: Success Score home with top actions, attendance planner, results and backlog tracking, engagement points (approved activities only), aptitude test, coding test with in-browser editor, soft skills, mock interview with voice and camera preview, feedback page.
- Admin: attendance sessions, result declaration, interventions, student directory with risk, skill gap, predictions, risk trends, segments, model page.

VISUAL

The six-box flow across the top; below it two small panels, "Model side" and "Platform side", listing three items each. Use one accent colour for the data-to-risk part and another for the platform part.

SPEAKER SCRIPT (ABOUT 160 WORDS)

> *"Our data is 800 synthetic students across five departments. We build features using only information available at the end of each semester, and we do the cleaning after that time cut, so there is no leakage. We prefer logistic regression for explainability and use boosting only when it clearly wins. Risk comes out as calibrated probabilities with two tiers. The platform runs on Supabase with row-level security on every table, so a mentor only sees their mentees, a head of department sees their department, and the placement officer sees everyone. The student portal and admin screens sit on top."*

Source: ml/README.md, supabase/migrations, lib/access.ts, student-portal/, app/admin.

Slide 5: Why This Approach Works

1:00 · Member C

> **Held-out results beat the Round 1 score on every outcome, and we state the limits.**

ON THE SLIDE: ONE TABLE (160 HELD-OUT STUDENTS)

Outcome	Model AUC	Round 1 score AUC	Recall
SGPA below 6	**0.905**	0.795	0.94
Backlog risk	**0.928**	0.885	0.82
Placement readiness	**0.939**	0.849	0.87
Engagement drop	**0.681**	0.436	0.80

THREE SUPPORTING FACTS

- **Risk spread at forecast:** 19.5% High, 32% Medium, 48% Low.
- **Does not just copy Round 1:** removing the Round 1 score changes AUC by less than 0.01.
- **Verified:** 79 security checks, 21 dataset checks, and model outputs that come out byte-identical when rerun.

WHY EACH DESIGN CHOICE WORKS

Choice	Reason
Cut by time before cleaning	Imputing first leaks future information
Recall target 0.85	Missing an at-risk student is the costly error; keeps held-out recall at or above 0.80
Two risk tiers	The recall threshold alone flags about a third of students
Drivers filtered for direction	Raw contributions looked contradictory because features are correlated
Logistic regression preferred	Mentors can read it; boosting must clearly win

Say these limits yourselves: the data is synthetic; engagement is the weakest model (about half of students flagged at recall 0.8); the backlog gain is in precision, not recall; the faculty rating is an inflated driver caused by the synthetic data.

VISUAL

The results table with the model column highlighted, and a small bar of the 19.5 / 32 / 48 risk spread in red, amber, green. Put a "Limits" ribbon at the bottom in a calm grey.

SPEAKER SCRIPT (ABOUT 130 WORDS)

"On 160 held-out students, our models beat the Round 1 score on all four outcomes. The biggest change is engagement drop, from 0.44 to 0.68. We tuned for recall because missing an at-risk student is the costly mistake. We also checked that the result does not just copy Round 1: removing that score barely changes anything. We want to be honest: the data is synthetic and engagement is still our weakest model, so we label it lower confidence."

Source: public/data/model-metrics.json, ml/README.md.

Slide 6: Impact, Feasibility & Next Steps

0:45 · Member C

> **It is built, tested and ready for real data; the next steps are clear and small.**

IMPACT (DELIVERED BY THE PRODUCT)

- Prioritised, filterable recommendations per student and per segment, each with an owner.
- Risk trends by department, year and segment for HODs and the Dean.
- Students see their own plan, eligibility rules and what-if outcomes.
- Executive brief page for the Dean's decision.

FEASIBILITY (ALREADY TRUE)

- Code is on GitHub (Raj2381/grow-track) with a README.
- Supabase schema with security policies and automated tests.
- Models, rules and what-if are reproducible with one command (npm run ml:all).
- Role sessions in the demo use real database accounts.

NEXT STEPS (THE PROJECT'S OWN OPEN ITEMS)

1. Replace synthetic data with real institute data and retrain.
2. Move the remaining shared admin pages from the bundled dataset to the database.
3. Turn on login enforcement (already built, currently off for the demo).
4. Add embedding-based relevance and the LLM grader to interview scoring (needs an API key).
5. Recheck the engagement model on real labels.

VISUAL

Three columns: "Delivered", "Ready", "Next", with a green tick on the first two and an arrow on the third.

SPEAKER SCRIPT (ABOUT 90 WORDS)

> *"The impact is a prioritised list of actions, each with an owner, plus risk trends for leaders and a clear plan for students. It is already built and tested, and everything is reproducible. The next steps are practical: load real institute data and retrain, move the remaining shared admin pages to the database, switch on login enforcement, and add the LLM grader to interview scoring."*

Source: README.md, supabase/, memory notes on open items. Do not add steps that are not in the project.

Slide 7: Why Our Solution Should Win

0:30 · Member A

> **It does not just report risk: it predicts, explains, recommends and tests what to change.**

ON THE SLIDE: THREE TILES MAPPED TO THE SCORING CRITERIA

Criterion	Our claim	Proof point
Innovation (35%)	Predict → explain → recommend → what-if in one product	1,027 owned recommendations; what-if matches the Python model
Business understanding (30%)	Built for four real roles and their decisions	Mentor, HOD, TPO and student workspaces with role-scoped data
Defence (20%)	Honest metrics and safe design	AUC 0.68 to 0.94 versus Round 1; leakage controls; row-level security

FINAL LINE (SAY IT, THEN STOP)

> "*GrowthTrack turns student data into the right action, for the right person, before it is too late.*"

DELIVERY TIPS

- No recap and no "thank you" slide of text; end on this slide and take questions.
- Look at the jury, not the screen. Pause after the final line.

Q&A defence sheet (3 minutes)

Answers use only what the project contains. Keep each to about 20 seconds; the team member who knows the topic answers.

Likely question	Answer
Is the data real?	No, it is synthetic: 800 students from nine archetypes, checked by 21 automated tests. The pipeline is built so real data can replace it.
Why is engagement only 0.68?	It is a noisy outcome. We still beat the old score (0.44) but flag it as lower confidence, and about half of students are flagged at recall 0.8.
How do you prevent leakage?	Features use only data up to the end of semester N; the time filter is applied before imputation; the dataset's diagnostic archetype column is never a feature.
Why logistic regression?	It is readable for mentors. Gradient boosting is used only if it beats it by 0.01 AUC; only engagement picks it, and we note that its drivers explain the logistic model.
How are recommendations made?	A rule table maps drivers to actions, with owner and priority scoring. They are tied to the student's own drivers, not generic.
What if the prediction is wrong?	It is decision support. Drivers are visible and a human mentor owns every action. Students cannot edit predictions.
How is data protected?	Row-level security on all tables, role-scoped views, anonymous faculty evaluations, secrets kept on the server.
Is login on?	The demo signs in automatically by role area. Enforcement is built and switched by a setting.
Do all pages use the database?	The role dashboards and the copilot do. Some shared admin pages still read the bundled dataset; moving them is next.
How does the interview scorer avoid rewarding junk?	Four stages: pre-checks (empty, gibberish, copied question, filler, self-introduction), a relevance gate, rubric scoring (keywords, concepts, clarity) with penalties for keyword stuffing and wrong claims, and an optional LLM check taking the lower score. Runs on the server.
Is an LLM required?	No. The copilot and interview LLM features need an API key; the rule-based scoring and the models work without one.
What is the Success Score?	The Round 1 weighted score (academic .25, attendance .20, LMS .15, placement .15, skills .10, engagement .10, faculty .05). Bands: 75 and above On Track, 50 to 74 Needs Attention, below 50 At Risk.

KEY NUMBERS TO MEMORISE

800 students · 5 departments · 15 tables · 4 outcomes · AUC .905 / .928 / .939 / .681 · 1,027 recommendations · 594 students covered · 6 segments · 27 silent decliners · 19.5% High / 32% Medium / 48% Low · 45 tables in the database · 79 security checks · 21 data checks